



Sri Lanka Institute of Information Technology

Web-Based Library Management System

Sequence Diagram

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Software Engineering | SE2030

MLB-B11G2-10

Group Members

IT25101420	Rayyan M.R.M
IT25102212	Vithanage C.B
IT25103166	Lakshitha P.M.M
IT25100467	Wijewickrama S.N
IT25102574	Wijegunawardana P.K.D.S.U
IT25100041	Amukotuwa A.G.S.I

Note on member workload distribution: The Requirement Gathering report defines ten major functions (Section 7.1–7.10) for a group of six members, so a strict one-function-per-member split is not possible. UC-01 to UC-04 and UC-06 each remain a single, substantial main function assigned to one member. Fine Management (7.2) and Reports and Analytics (7.6), taken alone, are narrower in scope than the other members' functions, so each of those two members has additionally been assigned the closely related extended functions from Section 7 to balance workload and ensure all ten major functions are covered: Vithanage C.B. covers Fine Management (UC-05), Fine Appeal and Waiver (UC-09), and Fine Payment and Receipt (UC-10) — the full fine-handling lifecycle; Amukotuwa A.G.S.I. covers Reports and Analytics (UC-06), Book Ratings and Reviews (UC-07), and Member Feedback (UC-08) — the system-insight and user-generated-content functions.

UC-01: Manage User Accounts and Roles

Member: Rayyan M.R.M. — IT25101420

Primary Actor:

Library Administrator

Supporting Actor:

Library Member, Librarian (for self-service registration and profile updates, per Requirements Section 7.1)

Goal:

Create, update, deactivate user accounts, and assign correct roles; and allow members and librarians to self-register and manage their own profile information.

Preconditions:

- The Library Administrator is logged in.
- The Library Administrator has permission to manage user accounts.
- For self-service actions, the Library Member or Librarian has a valid account or is registering a new one.

Main Flow – Administrator Manages Accounts:

- The Library Administrator selects “User Management”.
- The system displays the user-account list.
- The Library Administrator selects an existing user or chooses to create a new user.
- The Library Administrator enters or updates user information.
- The Library Administrator assigns a role such as Library Member, Librarian, Finance Officer, or Library Administrator.
- The Library Administrator saves the changes.
- The system validates the entered details.
- The system creates or updates the user account.
- The system displays a success message.

Main Flow – Self-Service Registration and Profile Update:

- The Library Member or Librarian selects “Register” or “My Profile”.
- The user enters or updates their personal information.
- The user saves the changes.
- The system validates the entered details.
- The system creates or updates the account within the permissions allowed for that role.
- The system displays a confirmation message.

Alternative Flows:

- If required information is missing, the system requests the user to complete it.
- If the username or email already exists, the system displays an error.
- If the Library Administrator deactivates an account, the system prevents that user from logging in.
- A Library Member or Librarian cannot change their own assigned role; only the Library Administrator can assign or change roles.

Postconditions:

- The user account is created, updated, or deactivated.
- The assigned role determines the user’s system access.

Sequence Diagram — UC-01: Manage User Accounts and Roles

[Insert sequence diagram here] UC-01: Manage User Accounts and Roles

UC-02: Manage Book Catalog and Inventory

Member: Lakshitha P.M.M. — IT25103166

Primary Actor:

Librarian

Supporting Actor:

Library Administrator (per Requirements Section 7.4)

Goal:

Add, update, search, and remove book records while maintaining availability information.

Preconditions:

- The Librarian is logged in.
- The Librarian has permission to manage book records.

Main Flow:

- The Librarian selects “Book Catalog and Inventory Management”.
- The system displays the current book catalog.
- The Librarian selects “Add New Book”.
- The Librarian enters the title, author, category, ISBN, number of copies, and other relevant details.
- The Librarian saves the book record.
- The system validates the data.
- The system adds the book to the catalog and sets the availability status.
- The system displays a confirmation message.

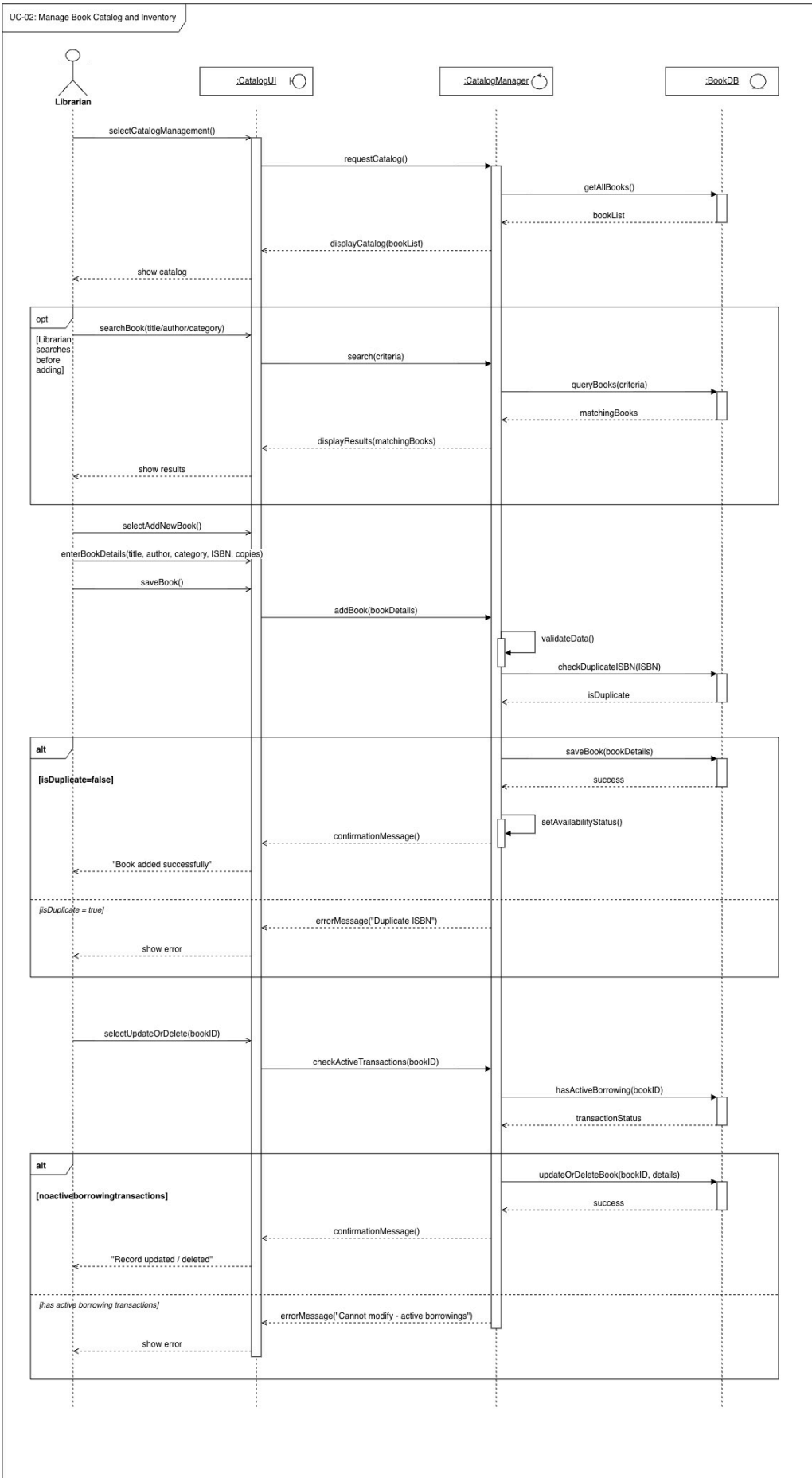
Alternative Flows:

- The Librarian may search for a book by title, author, or category before adding a new record.
- If a duplicate ISBN is entered, the system displays an error.
- The Librarian may update book details or delete a record that has no active borrowing transactions.
- The Library Administrator may also access Book Catalog and Inventory Management to oversee or correct catalog records.

Postconditions:

- The book catalog and inventory are updated.
- Members and guest visitors can search and view the latest book availability.

Sequence Diagram — UC-02: Manage Book Catalog and Inventory



UC-03: Borrow and Return Books

Member: Wijegunawardana P.K.D.S.U. — IT25102574

Primary Actor:

Librarian

Supporting Actor:

Library Member, Academic Staff Member (for renewal)

Goal:

Issue available books to members and record returned books.

Preconditions:

- The Librarian is logged in.
- The Library Member has an active account.
- The selected book is available for borrowing.
- The Library Member has not exceeded the borrowing limit and has no unpaid borrowing restriction.

Main Flow – Borrow Book:

- The Librarian selects “Issue Book”.
- The Librarian searches for and selects the Library Member.
- The system displays the Library Member’s borrowing eligibility.
- The Librarian searches for and selects an available book.
- The system checks book availability.
- The system calculates the due date according to library rules.
- The Librarian confirms the transaction.
- The system records the borrowing transaction.
- The system changes the book status to unavailable or reduces the available-copy count.
- The system displays the due date and confirmation.

Main Flow – Return Book:

- The Librarian selects “Return Book”.
- The Librarian enters or searches for the borrowing record.
- The system displays the borrowed book and due date.
- The Librarian confirms the return.
- The system calculates an overdue fine, if applicable.
- The system updates book availability.
- The system records the return transaction.
- The system displays a confirmation and fine amount, if applicable.

Main Flow – Renew Book:

- The Library Member selects “Renew Book”.
- The system displays the current borrowed book and due date.
- The system checks whether another member is waiting (reservation) for the same book.
- If no one is waiting, the system extends the due date according to library rules.
- The system updates the borrowing record.
- The system displays the new due date and confirmation.

Alternative Flows:

- If the book is unavailable, the system does not allow issuing it.
- If the Library Member has outstanding restrictions, the system prevents borrowing.
- If a fine exists after a return, the system sends the fine record to Fine Management.
- If another member has an active reservation on the book, the system does not allow renewal.

Postconditions:

- The borrowing or return transaction is recorded.
- Book availability and the Library Member’s borrowing history are updated.

Sequence Diagram — UC-03 (Borrow Book flow)

[Insert sequence diagram here] UC-03 (Borrow Book flow)

Sequence Diagram — UC-03 (Return Book flow)

[Insert sequence diagram here] UC-03 (Return Book flow)

Sequence Diagram — UC-03 (Renew Book flow)

[Insert sequence diagram here] UC-03 (Renew Book flow)

UC-04: Manage Book Reservations

Member: Wijewickrama S.N. — IT25100467

Primary Actor:

Academic Staff Member (a type of Library Member)

Supporting Actor:

Librarian

Goal:

Reserve an unavailable book and manage the reservation waiting list fairly.

Preconditions:

- The Academic Staff Member is logged in.
- The selected book is currently unavailable.
- The Academic Staff Member has an active account.

Main Flow:

- The Academic Staff Member searches for a book.
- The system displays that the book is unavailable.
- The Academic Staff Member selects “Reserve Book”.
- The system checks whether the Academic Staff Member already has an active reservation for the same book.
- The system adds the Academic Staff Member to the reservation waiting list.
- The system displays the reservation status and queue position.
- When a copy becomes available, the system notifies the next eligible Academic Staff Member in the waiting list.
- The Librarian confirms the reserved-book allocation when the Academic Staff Member collects it.

Alternative Flows:

- If the book is available, the system suggests borrowing it instead of reserving it.
- If the Academic Staff Member already reserved the same book, the system displays an error.
- The Academic Staff Member may cancel the reservation before allocation.
- If the Academic Staff Member does not collect the book within the allowed period, the system cancels the reservation and offers it to the next eligible person in the waiting list.

Postconditions:

- The reservation status and waiting list are updated.
- The next eligible person receives priority when the book becomes available.

Sequence Diagram — UC-04: Manage Book Reservations

[Insert sequence diagram here] UC-04: Manage Book Reservations

UC-05: Manage Fines and Fine Payments

Member: Vithanage C.B. — IT25102212

Primary Actor:

Finance Officer

Supporting Actor:

Librarian

Goal:

View overdue fines, verify automatically calculated fine amounts, record payments, and update payment status.

Preconditions:

- The Finance Officer is logged in.
- A Library Member has an overdue-book fine or an existing unpaid fine record.

Main Flow:

- The Finance Officer selects “Fine Management”.
- The system displays outstanding fine records.
- The Finance Officer searches for a Library Member or selects a fine record.
- The system displays borrowing details, overdue days, calculated fine amount, and payment status.
- The Finance Officer verifies the fine amount.
- The Library Member makes the payment.
- The Finance Officer records payment information.
- The system updates the payment status to paid or partially paid.
- The system stores the payment transaction.
- The system displays a payment confirmation.

Alternative Flows:

- If no unpaid fine exists, the system displays “No outstanding fines”.
- If the entered payment amount is less than the fine amount, the system records a partial payment and retains the remaining balance.
- If the calculation is incorrect, the Finance Officer refers the record for correction before accepting payment.

Postconditions:

- Fine and payment records are updated.
- The Library Member’s outstanding balance is accurate.

Sequence Diagram — UC-05: Manage Fines and Fine Payments

[Insert sequence diagram here] UC-05: Manage Fines and Fine Payments

UC-06: Manage Fine Appeal and Waiver

Member: Vithanage C.B. — IT25102212

Primary Actor:

Library Member

Supporting Actor:

Finance Officer, Library Administrator

Goal:

Allow members to submit a fine appeal with a reason and view its status, and allow administrators to approve or reject the appeal, reduce or waive the fine, and record the decision.

Preconditions:

- The Library Member is logged in.
- The Library Member has an existing unpaid or disputed fine.

Main Flow:

- The Library Member selects “Appeal Fine” for a specific fine record.
- The Library Member enters the reason for the appeal.
- The Library Member submits the appeal.
- The system records the appeal with a “Pending” status.
- The Library Administrator, in consultation with the Finance Officer, reviews the appeal.
- The Library Administrator approves or rejects the appeal.
- If approved, the system reduces or waives the fine amount.
- The system records the decision and any comments.
- The system updates the appeal and fine status.
- The system notifies the Library Member of the outcome.

Alternative Flows:

- If the appeal is rejected, the fine remains payable in full.
- The Finance Officer may refer a disputed calculation for correction before a decision is made.
- The Library Member may view the appeal status at any time.

Postconditions:

- The fine appeal record and decision are stored.

- The fine amount and status reflect the administrator's decision.

Sequence Diagram — UC-06: Manage Fine Appeal and Waiver

[Insert sequence diagram here] UC-06: Manage Fine Appeal and Waiver

UC-07: Manage Fine Payment and Receipt

Member: Vithanage C.B. — IT25102212

Primary Actor:

Library Member

Supporting Actor:

Finance Officer

Goal:

Allow members to make full or partial fine payments using a selected payment method, view remaining balance and payment history, and generate, view, or print a receipt.

Preconditions:

- The Library Member is logged in.
- The Library Member has an outstanding fine.

Main Flow:

- The Library Member selects "Pay Fine".
- The system displays the outstanding fine amount.
- The Library Member selects a payment method.
- The Library Member enters the payment amount (full or partial).
- The Library Member confirms the payment.
- The system processes the payment.
- The system calculates the remaining balance.
- The system updates the fine status to Paid or Partially Paid.
- The system generates a unique receipt.
- The system displays the receipt and updated balance to the Library Member.

Alternative Flows:

- The Finance Officer may record a payment on behalf of the Library Member (e.g., a cash payment at the counter).

- If the payment amount is less than the fine amount, the system records a partial payment and retains the remaining balance.
- The Library Member may view past payment history and reprint previous receipts at any time.

Postconditions:

- The payment and receipt records are stored.
- The fine balance and status are accurate and up to date.

Sequence Diagram — UC-07: Manage Fine Payment and Receipt

[Insert sequence diagram here] UC-07: Manage Fine Payment and Receipt

UC-08: Generate Reports and Analytics

Member: Amukotuwa A.G.S.I. — IT25100041

Primary Actor:

Library Administrator

Supporting Actor:

Librarian

Goal:

Generate reports to support library decisions.

Preconditions:

- The user is logged in as a Library Administrator or Librarian.
- The system contains relevant book, borrowing, reservation, user, and fine data.

Main Flow:

- The Library Administrator or Librarian selects “Reports and Analytics”.
- The system displays available report types.
- The user selects a report, such as borrowed books, overdue books, fines, reservations, users, or inventory.
- The user selects a date range or filtering criteria if required.
- The system retrieves relevant data.
- The system generates the selected report.
- The system displays the report with totals and relevant details.
- The user downloads or prints the report if needed.

Alternative Flows:

- If no records match the selected criteria, the system displays an empty-report message.
- If an invalid date range is entered, the system requests a valid range.
- The user may change report filters and regenerate the report.

Postconditions:

- The requested report is available for viewing, downloading, or printing.
- Management can use the report for operational and financial decisions.

Sequence Diagram — UC-08: Generate Reports and Analytics

[Insert sequence diagram here] UC-08: Generate Reports and Analytics

UC-09: Manage Book Ratings and Reviews

Member: Amukotuwa A.G.S.I. — IT25100041

Primary Actor:

Library Member

Supporting Actor:

Library Administrator

Goal:

Allow members to rate and review previously borrowed books, edit or delete their own reviews, and allow administrators to remove inappropriate reviews.

Preconditions:

- The Library Member is logged in.
- The Library Member has previously borrowed the book being reviewed.

Main Flow:

- The Library Member selects a previously borrowed book.
- The Library Member selects “Add Rating and Review”.
- The Library Member enters a rating and a written review.
- The Library Member submits the review.
- The system validates the review.
- The system saves the rating and review.
- The system displays the review on the book’s catalog page.

Alternative Flows:

- The Library Member may edit or delete their own review at any time.

- If the Library Member has not borrowed the book, the system does not allow submitting a review.
- The Library Administrator may remove a review that is inappropriate or violates guidelines.

Postconditions:

- The book's rating and review list is updated.
- Other members and guest visitors can view the updated ratings and reviews.

Sequence Diagram — UC-09: Manage Book Ratings and Reviews

[Insert sequence diagram here] UC-09: Manage Book Ratings and Reviews

UC-10: Manage Member Feedback

Member: Amukotuwa A.G.S.I. — IT25100041

Primary Actor:

Library Member

Supporting Actor:

Library Administrator

Goal:

Allow members to submit, view, and edit feedback about library services, and allow administrators to review, respond to, and manage feedback status.

Preconditions:

- The Library Member is logged in.

Main Flow:

- The Library Member selects "Submit Feedback".
- The Library Member enters feedback details.
- The Library Member submits the feedback.
- The system records the feedback with a "Pending" status.
- The Library Administrator reviews the submitted feedback.
- The Library Administrator responds to the feedback.
- The system updates the feedback status (e.g., Reviewed, Resolved).
- The system notifies the Library Member of the response.

Alternative Flows:

- The Library Member may view or edit their own feedback before it is reviewed.

- If required information is missing, the system requests the Library Member to complete it.
- The Library Administrator may mark feedback as low priority or close it without a response if not actionable.

Postconditions:

- The feedback record and its status are updated.
- The Library Member can view the current status and any administrator response.

Sequence Diagram — UC-10: Manage Member Feedback

[Insert sequence diagram here] UC-10: Manage Member Feedback